UEC631: 5G WIRELESS COMMUNICATION

L	T	P	Cr
3	0	0	3.0

Course Objective: This course will cover state-of-the-art topics in wireless networking and mobile computing. The objective of the course is to introduce students to recent advances in mobile networking and sensing, with an emphasis on practical design aspects of mobile systems.

Introduction to Wireless Communication Systems: History of Wireless Communication and Future Trends, Narrowband, Wideband, Ultra-Wideband Communication Systems, Description of 2G, 3G, 4G and Hybrid Communication Systems, Brief Introduction of Digital Modulation Techniques, Wireless channel impairments and mitigation techniques, Cognitive radio self-organizing networks and spectrum sharing

Cellular Concepts and System Design Fundamentals: Introduction to Cellular Concepts and Cellular System Design Fundamentals, Frequency Reuse, Cell size: Merits and demerits Channel Assignment Strategies, Handoff Strategies, Interference and System Capacity, Trunking and Grade of Service, Cell Splitting, Sectoring, Repeaters and Microcell Zone Concepts.

Mobile architectures: Convergence of mobile and fixed architectures: backhaul, fronthaul, midhaul and protocol convergence , LTE, LTE-A, LTE-A-PRO, Introduction to 5G, 5G Architecture, 5G Mobile Edge Computing, FOG computing, 5G Radio Access Technologies, Concept of New Radio (NR), mmWave Propagation, Principles of MIMO systems, Massive MIMO, Distributed MIMO, Programmability and Softwarization, Network Function Virtualization, (NFV), Software Defined networking (SDN), Role of NFV and SDN in 5G, 5G and IoT.

Wireless local area networks: IEEE: 802.11, 802,11a, 80.11b, 802.11g, 80211.n, HetNet and small cell deployments, Network Coding and Security, Optical networks for backbone, Visible Light communication

Future mobile networks: Drone networking, Multi-UAV networks, architectures and civilian applications, Communication challenges and protocols for micro UAVs, Connected and autonomous cars, Wireless technologies for Vehicle-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V) communications, Automotive surrounding sensing with GHz and THz signals

Familiarization with standards: IEEE 802.11, IEEE 802.16t-2025: Broadband Wireless,
IEEE 802.15.4-2024: Low rate network like IoT, IEEE 802.3 (Ethernet), LTE, 5G NR, IMT-2020

Course Learning Outcomes: After completion of this course, the students will be able to:
1. Understand the structure of current 4G cellular networks (including LTE) and the requirements of 5G cellular networks
2. Analyze the performance of IEEE 802.11 Wi-Fi.
3. Evaluate the basic foundations of wide variety of common wireless networking standards
4. Describe modern network architecture paradigms.

Text Books:
1. Wireless Communications: Principles and Practice, by Theodore S. Rappaport, PHI.
2. Wireless Networking Complete, by Pei Zheng et al., Morgan Kaufman

Reference Books:
1. 802.11n: A Survival Guide, by Matthew Gast, O'Reilly Media.
2. 802.11ac: A Survival Guide, by Matthew Gast, O'Reilly Media.

UEC707 : NETWORK VIRTUALIZATION AND SOFTWARE DEFINED NETWORKING

L	T	P	Cr
2	0	2	3.0

Course Objective:
To have a deep understanding of two important, emerging network technologies: Software Defined Networking (SDN) and Network Functions Virtualization (NFV). Use SDN emulator (Mininet) to set up and test network topologies.

Software Defined Network: History of programmable networks and Evolution of Software Defined Networking (SDN), IETF Forces, Active Networking. Separation of Control and Data Plane - Concepts, Advantages and Disadvantages, OpenFlow, protocol, 4D network architecture. Traditional Networking versus SDN.

Control & Data Plane: Overview, distributed and centralized control plane & data plane. Control plane: Existing SDN Controllers including Floodlight and Open Daylight projects. Customization of Control Plane: Switching and Firewall.
Data Plane**:** Software-based and Hardware-based; Programmable Network Hardware.

Network Virtualization: Concepts, Applications, Existing Network Virtualization Framework (VMWare and others), Network Functions Virtualization (NFV) and Software Defined Networks: Concepts, Implementation and Applications.

Network Programmability: Introduction, Northbound Application Programming Interface, Current Languages and Tools.

Data Center Networks: Packet, Optical and Wireless Architectures Network Topologies

Use Cases of SDNs: Data Centers, Internet Exchange Points, Backbone Networks, Home Networks, Traffic Engineering.

Familiarization with standards: IEEE 1916.1-2025: performance frameworks, metrics, and requirements for SDN and Network Functions Virtualization (NFV) infrastructures, IEEE 802.1Q: Deals with VLAN tagging, critical for network virtualization (segmentation).

Laboratory Work

1. Set up and get familiar with SDN emulator – Mininet and set up a virtual network.
2. Basic mininet operations.
3. Manually control the switch.
4. Move the rules to SDN controller.
5. Set different forwarding rules for each switch in the controller.

Course Learning Objectives (CLO)
The students will be able to:

1. Analyze the SDN architecture and advantages of programmable networks over traditional networks.
2. Analyse the impact of decoupled control plane and data plane on network.
3. Identify network virtualization components and their applications.
4. Design and implement networking problems using SDN-friendly network emulator.